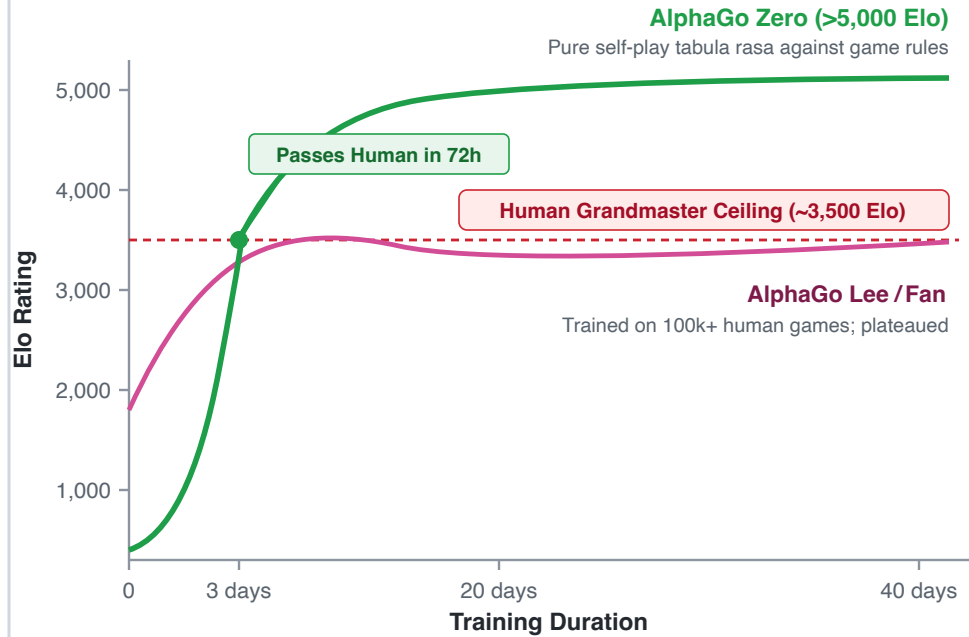


(a) Game AI: Demonstration Ceiling in Go

Silver et al. (Nature 2017)



Game Domain Characteristics:

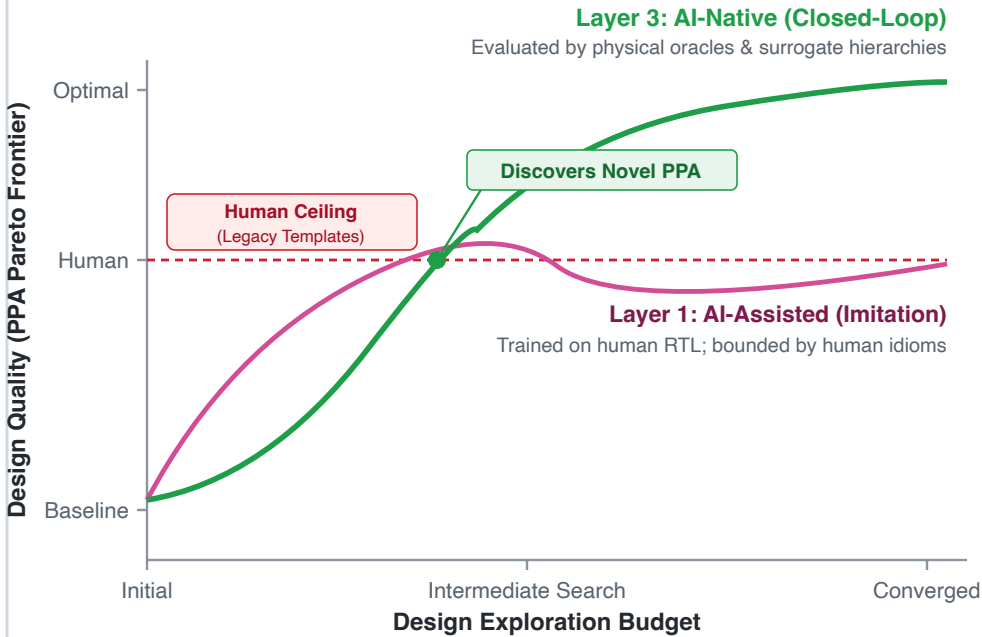
- **Evaluation Oracle:** Mathematical board rules (exact, deterministic)
- **Oracle Evaluation Cost:** $O(1)$ instantaneous, zero compute cost
- **Objective Function:** Scalar, binary outcome (win vs. loss)

Epistemic Takeaway: Human data acted as an anchor.

The system became superhuman only when human consensus was removed as the ceiling.

(b) Chip Architecture: Demonstration Ceiling in RTL

Architecture 2.0



Hardware Domain Realities:

- **Evaluation Oracle:** Physical signoff (STA, DRC, power, simulators)
- **Oracle Evaluation Cost:** Expensive (hours to days; demands multi-fidelity hierarchy)
- **Objective Function:** Multiobjective Pareto front (PPA, thermal, yield)

Physical Grounding: Reality arbitrates, not human style.

Closed-loop physical verification breaks through human cognitive templates.

Architectural Principle: Imitation of human demonstration corpora bounds exploration to human cognitive conventions.

Automated discovery of non-obvious microarchitectures requires evaluating candidates against empirical, physical ground truth.